FLAT PULLBACK OF THE DIVISOR OF A RATIONAL FUNCTION IS THE SUM OF THE DIVISORS ON THE IRREDUCIBLE COMPONENTS OF THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Dimension of the image of a cycle under a flat morphism. Same chapter chow.tex, label lemma-flat-inverse-image-dimension, invoked at proof line 3118. Gives dim_delta(Z') at least k - 1 for the closure Z' of the image of an integral Z of dimension k + r - 1, which is what reduces the proof to the two cases Z' = Y and Z' a prime divisor.
- Lengths and flat pullback of modules over a flat local homomorphism. Algebra, label algebra-lemma-pullback-module, invoked at proof line 3160. Identifies length_A(A/(x)) times length_B(B/m_A B) with length_B(B/xB), which is the form in which the desired equality is proved.
- Additivity of divisors under a restricted family. Same chapter chow.tex, label lemma-additivity-divisors-restricted, invoked at proof line 3169. Supplies the final equality expressing length_B(B/xB) as the sum over the minimal primes q_i of the residue length times the local length; the reduction to that equality is carried out in the counted span.

Source reference. Original lemma 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Navigation pointer for \ref{algebra-lemma-pullback-module} used inside the counted span (line 3160); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QG: sources/stacks-chow/pinned-original-source.tex, source lines 373–373. Navigation pointer for \ref{lemma-additivity-divisors-restricted} used inside the counted span (line 3169); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02R8: sources/stacks-chow/pinned-original-source.tex, source lines 2127–2127. Navigation pointer for \ref{lemma-flat-inverse-image-dimension} used inside the counted span (line 3118); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 3099); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y be schemes locally of finite type over S. Assume Y integral with $\dim_\delta(Y) = k$. Let $f : X \to Y$ be a flat morphism of relative dimension r. Then for $g \in R(Y)^*$ we have*

$$f^* div_Y(g) = \sum n_j i_{j,*} div_{X_j}(g \circ f|_{X_j})$$

as $(k + r - 1)$-cycles on X where the sum is over the irreducible components X_j of X and n_j is the multiplicity of X_j in X.

Proof. Let $Z \subset X$ be an integral closed subscheme of δ-dimension $k + r - 1$. We have to show that the coefficient n of $[Z]$ in $f^*\mathrm{div}(g)$ is equal to the coefficient m of $[Z]$ in $\sum i_{j,*}\mathrm{div}(g \circ f|_{X_j})$. Let Z' be the closure of $f(Z)$ which is an integral closed subscheme of Y. By Lemma 02R8 we have $\dim_\delta(Z') \geq k - 1$. Thus either $Z' = Y$ or Z' is a prime divisor on Y. If $Z' = Y$, then the coefficients n and m are both zero: this is clear for n by definition of f^* and follows for m because $g \circ f|_{X_j}$ is a unit in any point of X_j mapping to the generic point of Y. Hence we may assume that $Z' \subset Y$ is a prime divisor.

We are going to translate the equality of n and m into algebra. Namely, let $\xi' \in Z'$ and $\xi \in Z$ be the generic points. Set $A = \mathcal{O}_{Y,\xi'}$ and $B = \mathcal{O}_{X,\xi}$. Note that A, B are Noetherian, $A \to B$ is flat, local, A is a domain, and $\mathfrak{m}_A B$ is an ideal of definition of the local ring B. The rational function g is an element of the fraction field $Q(A)$ of A. By construction, the closed subschemes X_j which meet ξ correspond 1-to-1 with minimal primes

$$\mathfrak{q}_1, \ldots, \mathfrak{q}_s \subset B$$

The integers n_j are the corresponding lengths

$$n_i = \mathrm{length}_{B_{\mathfrak{q}_i}}(B_{\mathfrak{q}_i})$$

The rational functions $g \circ f|_{X_j}$ correspond to the image $g_i \in \kappa(\mathfrak{q}_i)^*$ of $g \in Q(A)$. Putting everything together we see that

$$n = \mathrm{ord}_A(g)\mathrm{length}_B(B/\mathfrak{m}_A B)$$

and that

$$m = \sum \mathrm{ord}_{B/\mathfrak{q}_i}(g_i)\mathrm{length}_{B_{\mathfrak{q}_i}}(B_{\mathfrak{q}_i})$$

Writing $g = x/y$ for some nonzero $x, y \in A$ we see that it suffices to prove

$$\mathrm{length}_A(A/(x))\mathrm{length}_B(B/\mathfrak{m}_A B) = \mathrm{length}_B(B/xB)$$

(equality uses Algebra, Lemma 02M1) equals

$$\sum\nolimits_{i=1,\ldots,s} \mathrm{length}_{B/\mathfrak{q}_i}(B/(x, \mathfrak{q}_i))\mathrm{length}_{B_{\mathfrak{q}_i}}(B_{\mathfrak{q}_i})$$

and similarly for y. As $A \to B$ is flat it follows that x is a nonzerodivisor in B. Hence the desired equality follows from Lemma 02QG. □